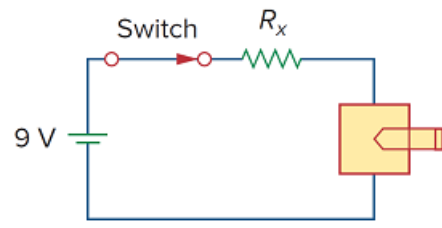


Problem

An electric pencil sharpener rated 240 mW, 6 V is connected to a 9-V battery as shown in Fig. 2.138. Calculate the value of the series-dropping resistor R_x needed to power the sharpener.

Figure 2.138:



Step-by-step solution

Step 1 of 3

Given electronic sharpener specifications

$$P_{\text{sharpener}} = 240\text{mW}$$

$$V_{\text{sharpener}} = 6\text{V}$$

Therefore internal resistance of the sharpener

$$R_{\text{sharpener}} = \frac{(V_{\text{sharpener}})^2}{P_{\text{sharpener}}}$$

$$R_{\text{sharpener}} = \frac{(6)^2}{240\text{m}}$$

$$R_{\text{sharpener}} = 150\Omega$$

Step 2 of 3

Current flowing through loop is given by

$$I = \frac{V_{\text{sharpener}}}{R_{\text{sharpener}}}$$

$$I = \frac{6}{150}$$

Voltage across unknown resistor R_x is

$$V_{R_x} = 9 - 6$$

$$V_{R_x} = 3\text{V}$$

Step 3 of 3

Now unknown resistor

$$R_x = \frac{V_{R_x}}{I}$$

$$R_x = \frac{3}{\frac{150}{6}}$$

$$R_x = \frac{3 \times 150}{6}$$

$$\boxed{R_x = 75\Omega}$$